

Using image recognition to automate assessment of cultural ecosystem services from social media photographs



Daniel R. Richards^{a,*}, Bige Tunçer^{a,b}

^aETH Zurich, Future Cities Laboratory, Singapore-ETH Centre, Singapore

^bArchitecture and Sustainable Design, Singapore University of Technology and Design, Singapore

ARTICLE INFO

Article history:

Received 8 March 2017

Received in revised form 2 September 2017

Accepted 6 September 2017

Available online 18 September 2017

Keywords:

Machine learning

Recreational ecosystem services

Singapore

Urban ecology

Recreation

ABSTRACT

Quantifying and mapping cultural ecosystem services is complex because of their intangibility. Data from social media, such as geo-tagged photographs, have been proposed for mapping cultural use or appreciation of ecosystems. However, manual content analysis and classification of large numbers of photographs is time consuming. This study develops a novel method for automating content analysis of social media photographs for ecosystem services assessment. The approach applies an online machine learning algorithm – Google Cloud Vision – to analyse over 20,000 photographs from Singapore, and uses hierarchical clustering to group these photographs. The accuracy of the classification was assessed by comparison with manual classification. Over 20% of photographs were taken of nature, being of animals or plants. The distribution of nature photographs was concentrated around particular natural attractions, and nature photographs were more likely to occur in parks and areas of high vegetation cover. The approach developed for clustering photographs was accurate and saved approximately 170 h of manual work. The method provides an indicator of cultural ecosystem services that can be applied rapidly over large areas. Automated assessment and mapping of cultural ecosystem services could be used to inform urban planning.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

Ecosystem services are the benefits that nature provides to people, including provisioning, regulating and cultural services (Millenium Ecosystem Assessment, 2003). To integrate ecosystem services within environmental decision making, we require information on a broad range of these services, so as to identify trade-offs among different objectives (Fish, 2011). Cultural ecosystem services are the non-material benefits that nature can provide, including recreational, spiritual and heritage values (Hernández-Morcillo et al., 2013). Quantifying cultural ecosystem services has traditionally been a time-consuming process, involving interviews (Plieninger et al., 2013), focus groups (Norton et al., 2012), or social surveys (Pleasant et al., 2014). More recently, data extracted from social media, such as geo-tagged photographs, have been used as indicators of recreational and aesthetic cultural ecosystem service value (Casalegno et al., 2013; Gliozzo et al., 2016; Keeler et al., 2015; Nahuelhual et al., 2013; Richards and Friess, 2015; Tenerelli et al., 2016; van Zanten et al., 2016; Wood et al., 2013). Data from social media offer great potential for quantifying ecosys-

tem services rapidly and over large areas, though for media such as photographs this still involves human supervision, making it a time-consuming process (Richards and Friess, 2015). This study presents an approach for automating the content analysis of social media photographs to facilitate rapid and large-scale ecosystem service mapping.

Previous studies that have used social media data in ecosystem service mapping have mapped the density of geo-tagged photographs as a proxy for public interest in an area (Casalegno et al., 2013; Keeler et al., 2015; van Zanten et al., 2016; Wood et al., 2013). Such studies have demonstrated that social-media data can be valuable in helping to map cultural ecosystem services over large areas, but there is a wealth of additional information in social media that is currently under-used. To extract more from the resource provided by social media, it is necessary to analyse the content of photographs, firstly to ensure that the photographs are relevant to the natural environment, and secondly to understand what aspects of the environment are of most interest to people in a particular area (Richards and Friess, 2015).

The density of photographs in a location corresponds closely to its popularity with visitors (Wood et al., 2013), but does not necessarily relate to public interest in the environment. The presence of a photograph does not tell us why people visited a location; for

* Corresponding author.

E-mail address: richards@arch.ethz.ch (D.R. Richards).

example, there may be a high density of photographs because of high public interest in a natural feature, or due to interest in a nearby popular restaurant. Photograph density as an index of cultural ecosystem service quality may be particularly confounded in urban areas, because recreational green spaces are found in close proximity to built infrastructure with high recreational potential, such as malls, cinemas, and restaurants. The risk of overestimating public interest in ecosystems can be reduced by discounting photographs that fall over non-natural areas (Gliozzo et al., 2016). However, this approach requires a complete knowledge of where those areas are, which may be difficult to obtain in heterogeneous urban-rural landscapes. In urban areas, many human-nature interactions occur in very small spaces, including community or private gardens (Marco et al., 2010; Seik, 2000). Such spaces may be hidden within larger areas that are assumed to be urban, and excluding all photographs from these areas could undervalue the importance of nature. The risk of overestimating public interest in the environment can also be limited by applying filters to the textual content (“tags”) associated with each image (van Zanten et al., 2016). However, photograph tagging relies on classification of the images by users, some of whom are less diligent in recording a broad range of tags, or may not record any tags at all.

There is a wealth of information held within photographs of the environment that can be analysed alongside social surveys and interviews to infer how, and why, people interact with nature (Dorwart et al., 2009). The density of nature photographs in a place can be considered as an indicator of public interest in nature in that area, which to some extent relates to recreational and cultural heritage ecosystem services (Richards and Friess, 2015). However, interpreting the information held within photographs can be a challenge for research into cultural uses of the environment, because the choice of what to take a photograph of is subjective on behalf of the photographer, and the motivation for taking the photograph may be unclear without additional contextual information (Scott and Canter, 1996). Despite the caveats in analysing the content of photographs, images from social media websites have previously been analysed by manual classification, to assess the relative importance of different cultural and recreational uses (Richards and Friess, 2015; Thiagarajah et al., 2015). Manual analysis of photographs can be consistent between assessors, and can be relatively rapid at small spatial scales; indeed, a sufficient analysis of one natural site can be completed in around 30 min (Richards and Friess, 2015). However, manual classification of photograph content does not scale up easily, as the time investment required to compare a large number of sites would be substantial. To allow rapid assessment of cultural ecosystem services over large areas, we require automated content analysis of photographs from social media.

The capability of image recognition software has improved rapidly over the past few years, spurred by the availability of large image datasets and high-power cloud computing (Agrawal et al., 2015; Kwak and An, 2016). Generally-applicable image recognition algorithms are now accessible online, such as Google’s Cloud Vision, and Microsoft’s Computer Vision (Google Cloud Vision, 2017; Microsoft Computer Vision, 2017). Google Cloud Vision, launched in 2016, provides an Application Programming Interface (API) for image recognition, allowing users to analyse individual images for their content, which is described in terms of keywords (Google Cloud Vision, 2017). Generalised image recognition APIs are now being applied to analyse the content of large number of images for research purposes, for example to categorise the content and gender balance of images in the global news (Kwak and An, 2016).

Automated content analysis of photographs from social media could help to ensure that only relevant photographs are included in indices of cultural ecosystem service quality, and may also pro-

vide more information, by distinguishing between different cultural ecosystem services, or aspects of the environment that are of interest (Di Minin et al., 2015; Richards and Friess, 2015). This study demonstrates a novel method for analysing photographs taken from social media with the Google Cloud Vision image recognition algorithm, and applies this approach across the city-state of Singapore. The objectives of the study were (1) to quantify the occurrence of photographs relating to nature, (2) to analyse the drivers of spatial variation in nature photographs, and (3) to assess the accuracy of the automated classification by comparison with a manual classification conducted by a human.

2. Method

2.1. Cultural ecosystem services in Singapore

Singapore is an island city-state of 5.6 million people, located in Southeast Asia (Department of Statistics, 2016). The population is relatively wealthy and there is a high rate of mobile phone ownership (IDA, 2014), making it a suitable country for analyses of social media data (Richards and Friess, 2015). While largely urban, around 56% of the land area is covered in managed and spontaneous vegetation, including public parks, gardens and nature reserves (Yee et al., 2011). Recreation is an important use of green space in Singapore, with a number of designated parks and gardens as well as networks of footpaths (Henderson, 2013; Tan, 2006).

2.2. Extraction of images and image recognition

Flickr is a photograph-sharing website with over 70 million users and 200 million geo-tagged photographs (Wood et al., 2013), which has become a commonly-used source of social-media photographs for assessing cultural ecosystem services (Richards and Friess, 2015; van Zanten et al., 2016; Wood et al., 2013). To download the photograph population across mainland Singapore, Sentosa Island, and Ubin Island, we laid out a regular grid of 165 locations and extracted all geo-tagged photographs from the social media website Flickr. The sample locations were arranged on a 2 km by 2 km grid. A random subset of 25000 photographs from Flickr (approximately 20% of the total) was then analysed further using image recognition. A random subset of all photographs was used, rather than stratifying the random sample by photographer. As such there is likely to be some bias in the data, as photographers that uploaded large numbers of photographs to Flickr were more likely to be represented. Each image was sent to the Google Cloud Vision API, which used a machine learning algorithm to assign keywords to the images (Google Cloud Vision, 2017). The Google Cloud Vision API was accessed through the RoogleVision package for the R statistical programming language (Teschner, 2016). A maximum of five keywords were returned for each image (Teschner, 2016).

2.3. Hierarchical clustering of images into general categories

The Google Cloud Vision image recognition algorithm returned up to five keywords that were associated with each photograph. A hierarchical clustering algorithm was applied to group the photographs according to their keywords (Oteros-rozas et al., 2017). A distance matrix was generated by comparing the proportion of the keywords assigned to one photograph that did not match the keywords assigned to another photograph. Hierarchical clustering was then applied using Ward’s distance, as implemented in the hclust function for the R statistical programming language (R Core Team, 2015). The appropriate number of clusters for analysis was assessed by plotting the average difference between the

within- and between- cluster variation for different numbers of clusters, and visually assessing the bend in the graph. The clusters identified by the hierarchical clustering were then used to categorise the photographs. To attach a meaning to each of the resulting clusters, the five most commonly-occurring words attributed to the photographs in each cluster were interpreted to subjectively define the type of photographs included in the groups.

2.4. Spatial modelling of photograph distributions

To understand the relationships between green spaces and the occurrence of nature photographs (those classed as “animal”, and “plant” photographs), photograph occurrence was modelled using maximum entropy modelling (MaxEnt). MaxEnt is commonly used to model species distributions from presence-only data (Elith et al., 2011). The probability of photograph occurrence was modelled as a function of four variables; (1) distance from the nearest major outdoor attraction, (2) the presence of parks including nature reserves, (3) the proportional coverage of forest within 50 m, and (4) the proportional coverage of managed vegetation within 0.01 km² grid squares. The 25 most popular outdoor attractions were identified using the tourist website Tripadvisor (www.tripadvisor.com), and details of these attractions can be found in Appendix S1. The area of parks and nature reserves was mapped partially using a dataset downloaded from the National Parks Board BIOME website (www.biome.nparks.gov.sg) in July 2014 that was modified manually to account for additional public open spaces that are not managed by the National Parks Board (Fig. 1). The proportion of forest and managed vegetation within 0.01 km² grid squares was quantified

using a 10 m resolution vegetation map derived from Sentinel-2 satellite imagery.

Vegetation was mapped using a supervised classification of radiometrically corrected (Level 1C) images from the Sentinel-2 satellite of the European Space Agency. Forest was defined as all unmanaged vegetation including scrub and forest, and managed vegetation was defined as turf and managed trees including street trees. Cloud cover is a significant issue in Southeast Asia, so seven separate images taken on different dates (between 8/12/2015 and 5/6/2016) were composited to ensure coverage over the whole study area. For each of the seven images, cloud cover was first classified, and cloudy areas were excluded from analysis. The land cover of the cloud-free areas was then classified using a random forest algorithm based on a training dataset of 309 point locations that were classified manually using Google Earth. The resulting process generated seven land cover maps from different occasions, each of which had missing data due to cloud cover. The seven maps were overlain, and the classified land cover at each pixel was defined as the modal land cover category (excluding cloud). The accuracy of the classification was initially assessed using an out-of-bag resampling process when building the random forest model, in which 80% of the dataset was used to train the model, and 20% was used to test the accuracy. The reported accuracy of the cloud cover classification was 89%, and the reported accuracy of the land cover classification was 81%. All classification and map processing was conducted in the R statistical language (R Core Team, 2015).

The quality of MaxEnt models is typically assessed using the area under the curve (AUC) of the receiver operating characteristic (ROC) plot. The AUC score can range between 1, indicating perfect prediction of the presence of photographs, and 0.5, indicating that

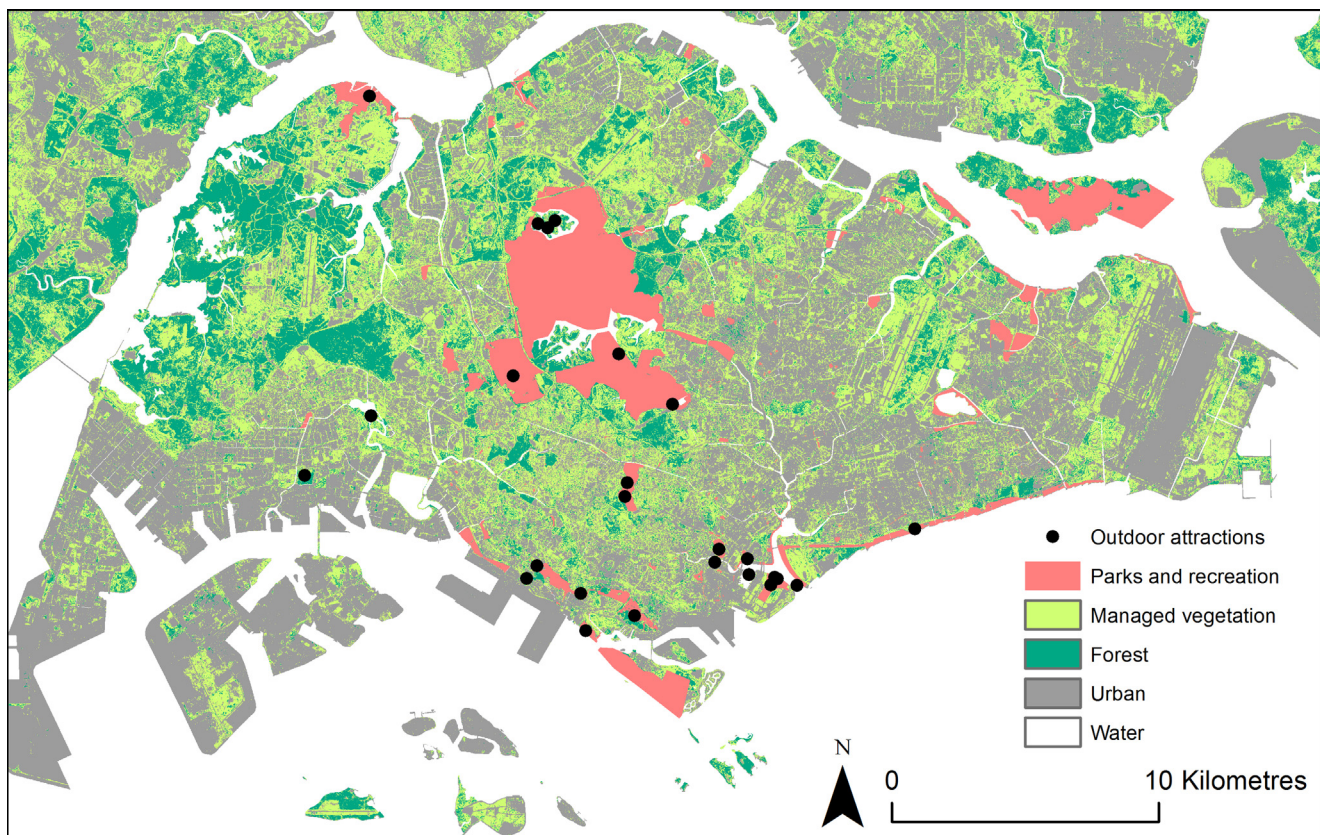


Fig. 1. Environmental data used to model the occurrence of plant and animal photographs. The proportional coverage of forest, and the proportional coverage of managed vegetation within 50 by 50 m grid squares, the presence of nature reserve, park, and recreation areas, and the distance from key outdoor attraction sites, was used to model the occurrence of plant and animal photographs using Maximum Entropy models. Land cover derived from modified Copernicus Sentinel data 2015–2016. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the model predictions are no better than random. MaxEnt models with AUC scores greater than 0.75 are generally considered acceptable (Phillips and Dudi, 2008). We trained the MaxEnt models using randomly sampled subsets (80%) of the datasets, and used the remaining 20% of the datapoints to test the predictive accuracy of the models. The AUC for the model run that includes the test dataset assesses the ability of the model to correctly predict new data. To visualise the effects of distance from focal points and habitat type on the occurrence of photographs, variable response curves for these factors were generated from the models for each type of photograph (Fig. 2).

2.5. Accuracy assessment of photograph classification

The accuracy of the combined image recognition and hierarchical clustering process was assessed quantitatively by comparing it

with a manual assessment conducted by a human. A randomly selected subset of 20 photographs from each photograph group was shown to an observer, who classified them manually into one of the seven categories given by the automated process. The percentage error was then compared using a confusion matrix. The agreement between the manual and automated classifications was further quantified using weighted Cohen's kappa, an index of agreement between different classifications commonly used in psychology (Landis and Koch, 1977).

3. Results

3.1. Extraction and clustering of photographs

The process extracted 130,115 unique photographs from Flickr, taken by 4174 photographers. The median number of photographs

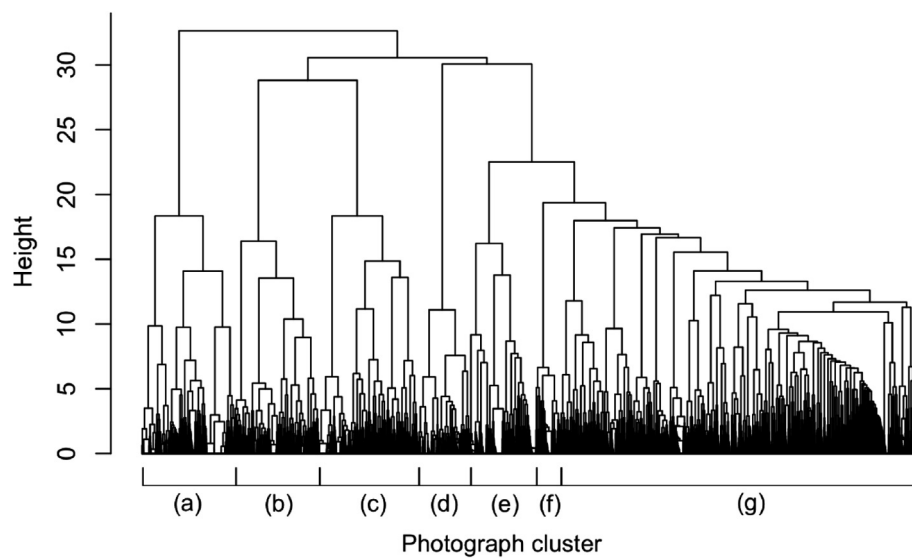


Fig. 2. Hierarchical clustering of photographs into seven groups. Clusters refer to (a) transport photographs, (b) plant photographs, (c) animal photographs, (d) food photographs, (e) photographs of people, (f) sports photographs, and (g) landscape and miscellaneous photographs.

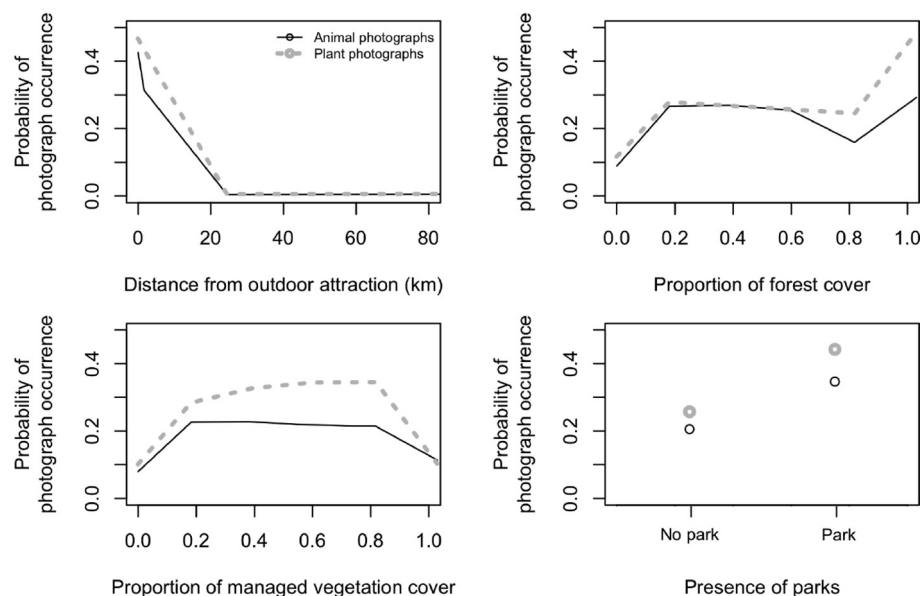


Fig. 3. Response curves from maximum entropy models for the four modelled environmental variables; distance from outdoor recreation attractions, proportion of forest cover within 50 m, proportion of managed vegetation within 50 m, and the location of parks including nature reserves.

taken by each photographer was 3, and 90% of photographers uploaded fewer than 20 photographs. The maximum number of photographs uploaded by a single photographer was 739. A random subset of 25,000 images were assigned 2200 unique keywords by the Google Cloud Vision algorithm. 1952 images were not assigned any keywords and were excluded from further analysis, leaving 23,048 images. Hierarchical clustering was applied to define seven major groups of photographs, which were were: transport ($n = 2768$), plant ($n = 2485$), animal ($n = 2951$), food ($n = 1540$), people ($n = 1951$), sports ($n = 727$), and landscape and miscellaneous photographs ($n = 10,626$). Miscellaneous photographs typically included blurry scenes, artistic photographs, and non-photographic images.

3.2. Spatial modelling of photograph distributions

The probability of occurrence of the plant and animal photographs was modelled separately. The AUC was high for both models; for the plant photographs the training AUC was 0.85 and the test AUC was 0.89, while for the social recreation photographs

the training AUC was 0.87 and the test AUC was 0.89. The distance from key outdoor attractions had the greatest percentage contribution to the plant photographs model (37%), followed by the proportion of forest cover (26%), presence of parks (26%), and proportion of managed vegetation cover (17%). The probability of occurrence of plant photographs decreased with increasing distance from outdoor attractions, was higher in parks, and increased with increasing forest and managed vegetation cover (Fig. 3). The distance from key outdoor attractions had the greatest percentage contribution to the animal photographs model (54%), followed by the proportion of forest cover (22%), the proportion of managed vegetation cover (17%) and the presence of parks (7%). The probability of occurrence of animal photographs decreased with increasing distance from outdoor attractions, was higher in parks, and increased with increasing forest and managed vegetation cover (Fig. 3). The modelled probability of occurrence of plant and animal photographs was broadly similar (Fig. 4a,b), with the probability of occurrence of animal photographs more tightly linked to the locations of outdoor attractions (Fig. 4b).

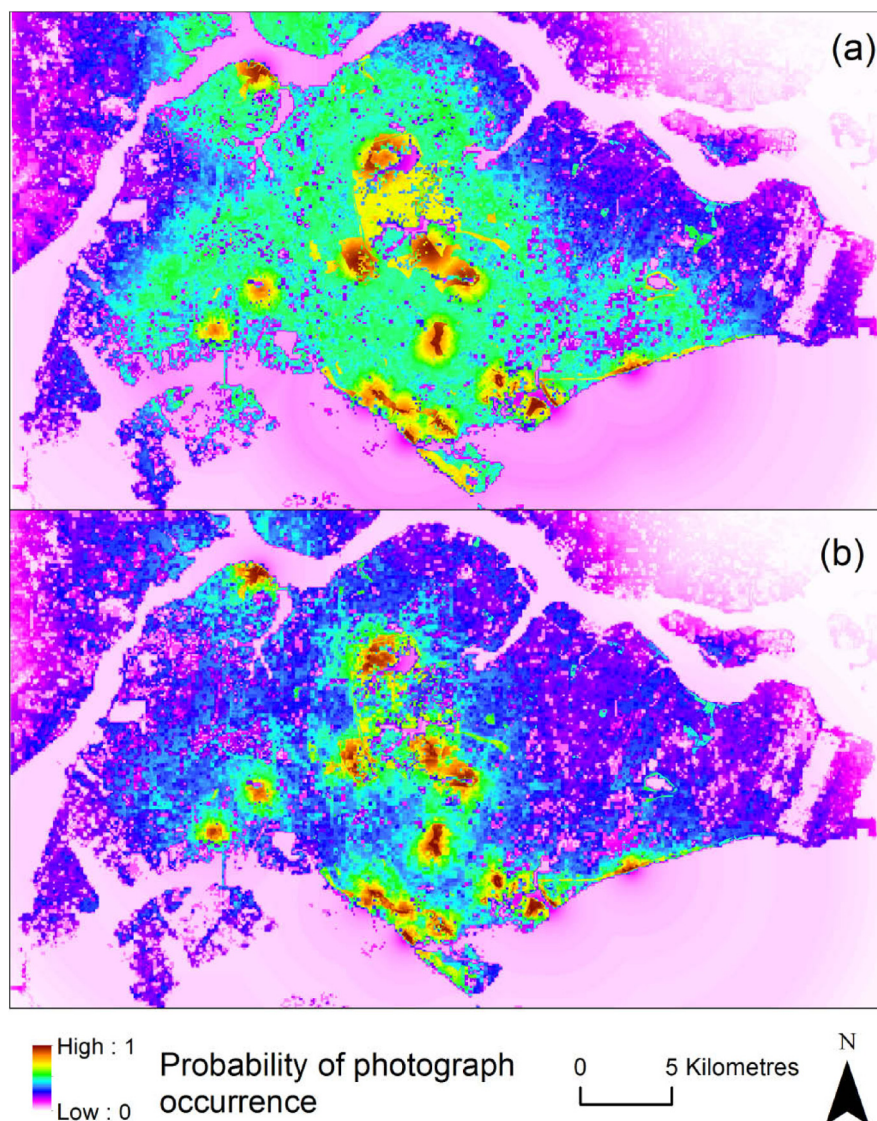


Fig. 4. Modelled probability of occurrence of (a) plant, and (b) animal photographs. White and pink areas indicate areas of low probability of photograph occurrence, yellow and brown areas indicate areas of high probability of photograph occurrence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3. Accuracy assessment of photograph classification

The overall accuracy of the photograph classification was 85%. The photograph group with the lowest accuracy was the landscape and miscellaneous category with an accuracy of 65% (Table 2). The accuracy of the plants category was 85% and the animals category was 95% (Table 2). The weighted Cohen's kappa of the agreement between the manual and automated classification was 0.83.

4. Discussion

4.1. Occurrence of nature in social media photographs in Singapore

Of the 23,048 photographs that were classified by Google Cloud Vision, over 20% were of animals or plants. Despite the high population and built density, there is an active and vocal nature conservation community present in Singapore (Kong et al., 1996; Wee and Hale, 2008), and nature tourism is popular at particular sites, including Sungei Buloh and the Central Catchment (Henderson, 2000). Nature photographs tended to focus on particular organismal groups, such as flowers, trees, and birds (Table 1), indicating that these taxa were either more common, more noticeable, or of greater interest to the photographer community (Richards et al., 2015). Birds are one of the most highly mobile and visible taxonomic groups present in cities, and the high occurrence of bird photographs within the “animal” category reflects findings from North America and globally that bird observation as a recreational activity attracts a substantial group of stakeholders with a keen interest in photographing this taxonomic group (Clucas et al., 2008; Nemésio et al., 2013).

Nature photographs were not taken uniformly across Singapore but were clustered around key locations (Fig. 4). Popular “focal points” for recreation and tourism exist because some locations offer particular attractions; either natural, such as the mangroves of the Sungei Buloh Wetland Nature Reserve (Henderson, 2000), or almost completely artificial, such as the Singapore Zoo and the Gardens by the Bay controlled environment planted areas (Davey, 2011). The popularity of natural and non-natural attractions is enhanced by advertising and personal recommendations, which can lead to increased visitor numbers (Kim, 2005). Concentration of recreational activity around key locations can have both positive and negative impacts on ecosystems. An advantage of concentrating recreational activity in particular areas is that it reduces visitor pressure elsewhere, providing space for species that are

sensitive to human activity (Burger et al., 2004). Concentration of tourism can also be good for tourists, as it allows infrastructure to be clustered to enhance efficiency and provide choice (Papatheodorou, 2004). Conversely, areas that experience high recreational use can become degraded, particularly when the characteristics of a place that attract visitors are natural and highly sensitive to humans (Müllner et al., 2004; Thompson et al., 2017).

Photographs of animals and plants were more likely to be taken in parks and nature reserves, possibly because animals and plants were more common in these areas than in urban areas. Nature reserves are typically designated due to their populations of wildlife and plants (Wee and Hale, 2008), and many parks are planted with aesthetically pleasing plant species (Khew et al., 2014). Nature photographs may also have been more likely to be taken in parks and nature reserves because these areas are publicly accessible, while access to many other forest patches is restricted (Richards and Friess, 2015). The occurrence of nature photographs increased with the proportional coverage of managed and forest vegetation, although this relationship plateaued, and even began to decrease at very high levels of managed vegetation (Fig. 3). Singapore has pursued an ambitious urban greening programme, with more than 50% of the land area now covered in vegetation (Yee et al., 2011). While substantial areas of forest are inaccessible as reserve or military land, large parts of the centre of the island are accessible as part of the MacRitchie, Central Catchment, and Bukit Timah Nature Reserves (Fig. 1).

The occurrence of both plant and animal photographs was best predicted by the distance from key outdoor attractions, but this relationship was stronger in the case of animal photographs. In contrast, plant photographs were more likely to occur in vegetated areas and parks outside the vicinity of key outdoor attractions (Fig. 4, Fig. 3a). Plants are typically easier to photograph than animals due to their stationary nature, and may also be relatively common as part of decorative planting, even within urban settings such as gardens and roadsides (Khew et al., 2014).

4.2. Automated processing of photographs for cultural ecosystem services assessment

The agreement between the manual and automated classification, as measured by the Cohen's kappa (Landis and Koch, 1977), was “substantial”, and comparable to the inter-rater agreement reported between multiple humans when classifying photographs from social media (Fleiss kappa of 0.75; Richards and Friess 2015).

Table 1
The top five most commonly occurring keyword tags in each of the seven photograph clusters.

Transport	Plants	Animals	Food	People	Sports	Landscape and misc.
Vehicle	Plant	Fauna	Food	Person	Sports	City
Transport	Tree	Vertebrate	Dish	People	Ball game	Art
Land vehicle	Flora	Wildlife	Meal	Youth	Team sport	Sky
Airplane	Flower	Bird	Cuisine	Child	Player	Night
Bus	Botany	Nature	Produce	Social group	Tournament	Color

Table 2
Confusion matrix of the accuracy of photograph classification. Accuracy assessed by comparison with 160 classified photographs.

	Transport	Landscape/misc.	Plants	Food	Animals	People	Sports	Accuracy
Food	17	2	0	0	0	1	0	85%
Landscape/misc.	2	13	2	0	0	3	0	65%
Transport	0	3	17	0	0	0	0	85%
People	0	1	1	17	1	0	0	85%
Plants	0	0	0	0	19	1	0	95%
Animal	0	1	0	0	0	19	0	95%
Vegetation	0	0	0	0	0	3	17	85%

The large landscape and miscellaneous category reflects a challenge in defining photographs based on a limited amount of information (only five keywords), when there is a relatively large range of possible keywords. Photographs that were assigned relatively specific keywords were thus difficult to classify and fell into the miscellaneous category, even if their content was relevant to another category. Improvements in the clustering of extracted images could be made by developing and applying thesaurus algorithms that can identify and match keywords with similar or related meanings (Roussinov, 1998). The extraction of keywords from social media feeds has been investigated, with the purpose of understanding which topics of conversations various places generate (You and Tunçer, 2016). A similar extraction and clustering approach could be used to classify images.

While there was some classification error in the content analysis of the social media photographs, the automated nature of the assessment presented here gives it great potential for rapidly scaling up cultural ecosystem service assessment over large areas. If the content analysis had been conducted following the manual method described in Richards and Friess (2015), it would have taken approximately 170 h to analyse the number of photographs presented in the current study. Such a large time investment would present a major barrier to conducting a similar mapping exercise using a manual content analysis, and it would be impractical to expand the analysis to cover a wider area and number of photographs. Despite the advantages of automated content analysis of photographs, there are some benefits to manual viewing that may make it preferable in some circumstances; a human classifier can obtain more information from a photograph, such as the species of animal present, or the cultural context in which a photograph of people was taken. A hybrid approach, with automated content analysis supplemented by viewing of a small number of photographs by an expert, may provide additional qualitative information that is useful for ecosystem service management.

The automated approach to image classification applied in this study is cost-effective when compared to manual classification of images, but is not free. At the time of writing (July 2017), the first 1000 images sent to the Google Cloud Vision API within a month are freely analysed, and the analysis of each 1000 images after that costs \$1.50 (US Dollars). While not a huge expense, the cost of analysing images using Google Cloud Vision may constrain the use of the approach at very large (i.e. global) scales. Alternative online image recognition APIs, such as the Microsoft Computer Vision API, may provide a more cost-effective option in future (Microsoft Computer Vision, 2016), or simpler purpose-built image recognition algorithms could be trained to identify natural content (Saitoh and Kaneko, 2003).

4.3. Limitations of using social media photographs to assess cultural ecosystem services

The occurrence and density of photographs of nature can provide an indicator of public interest in nature in that location, but there is a disconnect between such an indicator and a measure of cultural ecosystem service value. The motivations for people to take photographs of nature varies; in some cases, people take photographs to record positive attributes of the environment that they find appealing, while other photographs are taken to record negative attributes of the environment (Dorwart et al., 2009). Furthermore, photographs can be taken to represent a place as it appears as a physical object, or interpreted through the lens of a person's memories and experiences surrounding a place (Scott and Canter, 1996). It is therefore complex to attach a specific cultural ecosystem service value to the indicator presented here; people may take photographs of nature at a location while they are using it for recreation, because they are creating art, or because

they are documenting what they see as an important natural heritage. Content analysis of social media photographs to evaluate cultural ecosystem services should therefore be aware of the uncertainty surrounding photograph content. In our approach, we consider the occurrence of nature photographs as general indicator of the public interest in nature. To understand more clearly why people take photographs in a particular place, and which cultural ecosystem services are represented, additional information on the context of the photograph may be required. It may be possible to gain context on the use of green spaces through the meta-data that is sometimes attached to social media photographs, such as the title, notes, comments and tags attached by users (Hollenstein and Purves, 2010). Alternatively, interviews or surveys with people at a location may provide additional context on the cultural ecosystem services that are most valued (Pleasant et al., 2014).

Any analysis of social media data is limited by the biases inherent in the source dataset, particularly as social media are most commonly used by younger people, and different demographic groups use different platforms (Pew Research Center's Internet & American Life, 2013; van Zanten et al., 2016). Analysis of social media photographs should therefore not be the only approach applied to quantify cultural ecosystem services, but can be a useful tool to provide quantitative data over large spatial scales, that can supplement more in-depth qualitative analyses (Richards and Friess, 2015; Thiagarajah et al., 2015).

5. Conclusions

This study presents a method for using a publicly-available image recognition algorithm to interpret the content of a large number of images taken from social media, in order to quantify and map interest in nature. Such an approach can be used to map the predominate areas for nature photography, and distinguish between different types of nature photographs, as shown through the case study of Singapore. The approach developed for clustering photographs based on their content was accurate, and the method allows an indicator of cultural ecosystem services to be mapped rapidly over large areas. Automated assessment and mapping of cultural ecosystem services could help to capture some of the complex value of these services, and could be used to inform urban and regional planning of land uses. The method presented here joins a growing suite of automated approaches for quantifying ecosystem services (Dobbs et al., 2014; Richards and Edwards, 2017), which could help to fill the gaps in ecosystem service knowledge that remain prevalent, especially in the tropics (Song et al., 2017).

Acknowledgement

The research was conducted at the Future Cities Laboratory at the Singapore-ETH Centre, which was established collaboratively between ETH Zurich and Singapore's National Research Foundation (FI 370074016) under its Campus for Research Excellence and Technological Enterprise programme. The authors would like to thank Prof. Peter Edwards and two anonymous reviewers for their helpful additions to the manuscript.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ecoser.2017.09.004>.

References

- Agrawal, H., Mathialagan, C.S., Goyal, Y., Chavali, N., Banik, P., Mohapatra, A., Osman, A., Batra, D., 2015. Cloudcv: large-scale distributed computer vision as a cloud service. *Mob. Cloud Vis. Media Comput. From Interact. to Serv.* 265–290. http://dx.doi.org/10.1007/978-3-319-24702-1_11.
- Burger, J., Jeitner, C., Clark, K., Niles, L.J., 2004. The effect of human activities on migrant shorebirds: successful adaptive management. *Environ. Conserv.* 31, 283–288.
- Casalegno, S., Inger, R., Desilvey, C., Gaston, K.J., 2013. Spatial covariance between aesthetic value & other ecosystem services. *PLoS One* 8, e68437. <http://dx.doi.org/10.1371/journal.pone.0068437>.
- Clucas, B., McHugh, K., Caro, T., 2008. Flagship species on covers of US conservation and nature magazines. *Biodivers. Conserv.* 17, 1517–1528. <http://dx.doi.org/10.1007/s10531-008-9361-0>.
- Davey, M., 2011. Gardens by the bay: Ecologically reflective design. *Archit. Des.* 81, 108–111.
- Department of Statistics, S., 2016. Population Trends, 2016 156.
- Di Minin, E., Tenkanen, H., Toivonen, T., 2015. Prospects and challenges for social media data in conservation science. *Front. Environ. Sci.* 3, 1–6. <http://dx.doi.org/10.3389/fenvs.2015.00063>.
- Dobbs, C., Nitschke, C.R., Kendal, D., 2014. Global drivers and tradeoffs of three urban vegetation ecosystem services. *PLoS One* 9 (2014), e113000. <http://dx.doi.org/10.1371/journal.pone.0113000>.
- Dorwart, C.E., Moore, R.L., Leung, Y.-F., 2009. Visitors' perceptions of a trail environment and effects on experiences: a model for nature-based recreation experiences. *Leis. Sci.* 32, 33–54. <http://dx.doi.org/10.1080/01490400903430863>.
- Elith, J., Phillips, S.J., Hastie, T., Dudík, M., Chee, Y.E., Yates, C.J., 2011. A statistical explanation of MaxEnt for ecologists. *Divers. Distrib.* 17, 43–57.
- Fish, R.D., 2011. Environmental decision making and an ecosystems approach: some challenges from the perspective of social science. *Prog. Phys. Geogr.* 35, 671–680. <http://dx.doi.org/10.1177/0309133311420941>.
- Giozzio, G., Pettorelli, N., Haklay, M., 2016. Using crowdsourced imagery to detect cultural ecosystem services: a case study in South Wales, UK. *Ecol. Soc.* 21, art6. <http://dx.doi.org/10.5751/ES-08436-210306>.
- Google Cloud Vision, 2017. Documentation for the Google Cloud Vision API. Available online: www.cloud.google.com/vision/ Accessed 01/07/2017.
- Henderson, J.C., 2013. Urban parks and green spaces in Singapore. *Manag. Leis.* 18, 213–225. <http://dx.doi.org/10.1080/13606719.2013.796181>.
- Henderson, J.C., 2000. Managing tourism in small islands: the case of Pulau Ubin. *Singapore J. Sustain. Tour.* 8, 250–262. <http://dx.doi.org/10.1080/09669580008667361>.
- Hernández-Morcillo, M., Plieninger, T., Bieling, C., 2013. An empirical review of cultural ecosystem service indicators. *Ecol. Indic.* 29, 434–444. <http://dx.doi.org/10.1016/j.ecolind.2013.01.013>.
- Hollenstein, L., Purves, R., 2010. Exploring place through user-generated content: Using Flickr to describe city cores. *Jsis* 1, 21–48.
- IDA, 2014. Statistics on Telecom Services for 2012 (July–December). Infocomm Development Authority of Singapore, URL <http://www.ida.gov.sg/Infocomm-Landscape/Facts-and-Figures/Telecommunications/Statistics-on-Telecom-Services/>.
- Keeler, B.L., Wood, S.A., Polasky, S., Kling, C., Filstrup, C.T., Downing, J.A., Keeler, B.L., Wood, S.A., Polasky, S., Kling, C., Filstrup, C.T., Downing, J.A., 2015. Recreational demand for clean water: evidence from geotagged photographs by visitors to lakes. *Front. Ecol. Environ.* 13, 76–81.
- Khew, J.Y.T., Yokohari, M., Tanaka, T., 2014. Public perceptions of nature and landscape preference in Singapore. *Hum. Ecol.* 42, 979–988.
- Kim, D.-Y., 2005. Modeling Tourism Advertising Effectiveness. *J. Travel Res.* 44, 42–49.
- Kong, L., Yeoh, B.S.A., Wallace, A.R., 1996. Social constructions of nature in urban Singapore. *Southeast Asian Studies* 34, 402–423.
- Kwak, H., An, J., 2016. Revealing the hidden patterns of news photos: analysis of millions of news photos using GDELT and deep learning-based vision. The Workshops of the Tenth International AAAI Conference on Web and Social Media News and Public Opinion: Technical Report WS-16-18, 99–07.
- Landis, J.R., Koch, G.G., 1977. The measurement of observer agreement for categorical data. *Biometrics* 33, 159–174. <http://dx.doi.org/10.2307/2529310>.
- Marco, A., Barthelemy, C., Dutoit, T., Bertaudière-Montes, V., 2010. Bridging human and natural sciences for a better understanding of urban floral patterns: the role of planting practices in Mediterranean gardens. *Ecol. Soc.* 15, 4. doi:citeulike-article-id:8040972.
- Microsoft Computer Vision, 2017. Documentation for the Microsoft Computer Vision API. Available online: www.microsoft.com/cognitive-services/en-us/computer-vision-api, accessed 01/07/2017.
- Millennium Ecosystem Assessment, 2003. Ecosystems and human well-being: a framework for assessment. World resources institute, Washington, DC 5.
- Müllner, A., Eduard Linsenmair, K., Wikelski, M., 2004. Exposure to ecotourism reduces survival and affects stress response in hoatzin chicks (*Opisthocomus hoazin*). *Biol. Conserv.* 118, 549–558.
- Nahuelhual, L., Carmona, A., Lozada, P., Jaramillo, A., Aguayo, M., 2013. Mapping recreation and ecotourism as a cultural ecosystem service: an application at the local level in Southern Chile. *Appl. Geogr.* 40, 71–82. <http://dx.doi.org/10.1016/j.apgeog.2012.12.004>.
- Nemésio, A., Seixas, D.P., Vasconcelos, H.L., 2013. The public perception of animal diversity: What do postage stamps tell us? *Front. Ecol. Environ.* 11, 9–10. <http://dx.doi.org/10.1890/13.WB.001>.
- Norton, L.R., Inwood, H., Crowe, a., Baker, a., 2012. Trialling a method to quantify the “cultural services” of the English landscape using Countryside Survey data. *Land Use Policy* 29, 449–455. <http://dx.doi.org/10.1016/j.landusepol.2011.09.002>.
- Oteros-rozas, E., Martín-lópez, B., Fagerholm, N., Bieling, C., Plieninger, T., 2017. Using social media photographs to explore the relation between cultural ecosystem services and landscape features across five European sites. *Ecol. Indic.* <http://dx.doi.org/10.1016/j.ecolind.2017.02.009>.
- Papathodorou, A., 2004. Exploring the evolution of tourism resorts. *Ann. Tour. Res.* 31, 219–237.
- Pew Research Center's Internet & American Life, 2013. The Demographics of Social Media Users – 2012, 1–14.
- Phillips, S.J., Dudík, M., 2008. Modeling of species distributions with Maxent: new extensions and a comprehensive evaluation, 161–175.
- Pleasant, M.M., Gray, S.A., Lepczyk, C., Fernandes, A., Hunter, N., Ford, D., 2014. Managing cultural ecosystem services. *Ecosyst. Serv.* 1–7. <http://dx.doi.org/10.1016/j.ecoser.2014.03.006>.
- Plieninger, T., Dijk, S., Oteros-Rozas, E., Bieling, C., 2013. Assessing, mapping, and quantifying cultural ecosystem services at community level. *Land Use Policy* 33, 118–129. <http://dx.doi.org/10.1016/j.landusepol.2012.12.013>.
- R Core Team, 2015. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. Available online: <https://www.R-project.org/>.
- Richards, D.R., Edwards, P.J., 2017. Quantifying street tree regulating ecosystem services using Google Street View. *Ecol. Indic.* 77, 31–40. <http://dx.doi.org/10.1016/j.ecolind.2017.01.028>.
- Richards, D.R., Friess, D.A., 2015. A rapid indicator of cultural ecosystem service usage at a fine spatial scale: content analysis of social media photographs. *Ecol. Indic.* 53, 187–195.
- Richards, D.R., Warren, P.H., Moggridge, H.L., Maltby, L., 2015. Spatial variation in the impact of dragon flies and debris on recreational ecosystem services in a floodplain wetland 15, 113–121.
- Roussinov, Dmitri G., Hsinchun Chen, 1998. A scalable self-organizing map algorithm for textual classification: A neural network approach to thesaurus generation. *Commun. Cognition Artificial Intell.* J.
- Saitoh, T., Kaneko, T., 2003. Automatic recognition of wild flowers. *Syst. Comput. Japan* 34, 90–101.
- Scott, M.J., Canter, D.V., 1996. Picture or place? A multiple sorting study of landscape. *J. Environ. Psychol.* 17, 263–281.
- Seik, F.T., 2000. Experiences from a community nature project in a housing estate in Singapore. *Local Environ.* 5, 285–297. <http://dx.doi.org/10.1080/13549830050134239>.
- Song, X.P., Richards, D.R., Edwards, P.J., Tan, P.Y., 2017. Benefits of trees in tropical cities. *Science* 356, 6344.
- Tan, K.W., 2006. A greenway network for Singapore. *Landsc. Urban Plan.* 76, 45–66. <http://dx.doi.org/10.1016/j.landurbplan.2004.09.040>.
- Teschner, F., 2016. RoogVision: Access to Google's Cloud Vision API for Image Recognition, OCR and Labeling. R package version 0.0.1.1.
- Tenerelli, P., Demšar, U., Luque, S., 2016. Crowd sourcing indicators for cultural ecosystem services: a geographically weighted approach for mountain landscapes. *Ecol. Indic.* 64, 237–248. <http://dx.doi.org/10.1016/j.ecolind.2015.12.042>.
- Thiagarajah, J., Wong, S.K.M., Richards, D.R., Friess, D.A., 2015. Historical and contemporary cultural ecosystem service values in the rapidly urbanizing city state of Singapore. *Ambio* 44, 666–667. <http://dx.doi.org/10.1007/s13280-015-0647-7>.
- Thompson, B.S., Gillen, J., Friess, D.A., 2017. Challenging the principles of ecotourism: insights from entrepreneurs on environmental and economic sustainability in Langkawi, Malaysia. *J. Sustain. Tour.* <http://dx.doi.org/10.1080/09669582.2017.1343338>.
- van Zanten, B.T., van Berkel, D.B., Meetemeyer, R.K., Smith, J.W., Tieskens, K.F., Vergur, P.H., 2016. Continental scale quantification of landscape values using social media data. *Proc. Natl. Acad. Sci.* 113, 1–7.
- Wee, Y.C., Hale, R., 2008. The Nature Society (Singapore) and the struggle to conserve Singapore's nature areas. *Nat. Singapore* 1, 41–49.
- Wood, S.A., Guerry, A.D., Silver, J.M., Lacayo, M., 2013. Using social media to quantify nature-based tourism and recreation. *Sci. Rep.* 3, 2976. <http://dx.doi.org/10.1038/srep02976>.
- Yee, A.T.K., Corlett, R.T., Liew, S.C., Tan, H.T.W., 2011. The vegetation of Singapore – an updated map. *Gard. Bull. Singapore* 63, 205–212.
- You, L., Tunçer, B., 2016. Exploring the utilization of places through a scalable “Activities in Places” analysis mechanism. The 2016 IEEE International Conference on Big Data, IEEE BigData 2016, December 2016, Washington USA.